
N Stemi ST
Stemi ST
Sick Horses Only
Ddx
STRESS TEST
NITRO
NITRO
NITRO
DO NOT overFILL
PRO-grade NITRO 3Xplo5ive
GRANOLA
GRANOLA
Calci-Yum
NO OUTSIDE FOOD OR DRINKS
ASA
RISK
V1
V5
85x (220 - age)

1.2 - Stable Angina and Prinzmetal -or- Vasospastic Angina SOAP

1) Anvil:
Angina = chest discomfort caused by myocardial ischemia (imbalance between myocardial oxygen supply & demand)

2) Blacksmith clutching chest in pain:
Angina = **chest discomfort** caused by myocardial ischemia (i.e., imbalance between myocardial oxygen supply & demand)

9(S) Heavy anvil:
Patients with angina classically describe **chest heaviness, pressure, or tightness,** rather than sharp pain **(Chief Complaint)**

10(S) Blacksmith clenching fist over chest:
Levine's sign = **holding a clenched fist over the chest** when describing chest pain → frequently associated with ischemic chest pain **(HPI)**

11(S) Diffuse chest plate:
Ischemic chest pain is typically **diffuse and poorly localized (HPI)**

12(S) Tight leather brace on left arm:
Ischemic chest pain often **radiates to the left arm, neck, or jaw (HPI)**

15(S) Blacksmith swinging hammer:
Angina triggers = **exertion**, exercise, emotional stress, cold temperature, sex, drugs (e.g., cocaine, amphetamines) **(HPI)**

19(S) Blacksmith blowing puff of air:
Other symptoms associated with ischemic chest pain include **dyspnea,** diaphoresis, & nausea **(HPI)**

20(S) Assistant sweating:
Other symptoms associated with ischemic chest pain include dyspnea, **diaphoresis**, & nausea **(HPI)**

21(S) Nauseous assistant:
Other symptoms associated with ischemic chest pain include dyspnea, diaphoresis, & **nausea (HPI)**

22(S) Elderly woman blacksmith with candy hair tie:
Elderly, diabetic, & female patients with myocardial ischemia may present with atypical symptoms such as dyspnea, palpitations, or syncope) **(HPI)**

23(S) Elderly woman blowing puff of air:
Elderly, diabetic, & female patients with myocardial ischemia may present with atypical symptoms such as **dyspnea**, palpitations, or syncope) **(HPI)**

24(S) Vibrating heart sign on anvil:
Elderly, diabetic, & female patients with myocardial ischemia may present with atypical symptoms such as dyspnea, **palpitations**, or syncope) **(HPI)**

25(S) Sinking angina anvil:
Elderly, diabetic, & female patients with myocardial ischemia may present with atypical symptoms such as dyspnea, palpitations, or **syncope) (HPI)**

26(S) Old, male blacksmith:
Increased age (>55) and male sex increase risk of ischemic heart disease **(HPI)**

27(S) Obese stable hand:
Conditions that increase risk of ischemic heart disease include **obesity (BMI >30)**, diabetes, hyperlipidemia, hypertension, & CKD **(PMHx)**

28(S) Jar of Diasweeties candy:
Conditions that increase risk of ischemic heart disease include obesity (BMI >30), **diabetes**, hyperlipidemia, hypertension, & CKD **(PMHx)**

29(S) Bread & stick of butter:
Conditions that increase risk of ischemic heart disease include obesity (BMI >30), diabetes, **hyperlipidemia**, hypertension, & CKD **(PMHx)**

30(S) Red pipe with steam bursting out:
Conditions that increase risk of ischemic heart disease include obesity (BMI >30), diabetes, hyperlipidemia, **hypertension**, & CKD **(PMHx)**

31(S) Rusty kidney-shaped pulleys:
Conditions that increase risk of ischemic heart disease include obesity (BMI >30), diabetes, hyperlipidemia, hypertension, & **CKD (PMHx)**

32(S) Black paint stroke:
A history of atherosclerotic disease (e.g., **stroke**, peripheral arterial disease, ACS) increases risk of ischemic heart disease **(PMHx)**

33(S) Yellow lipid stains on sleeve:
A history of atherosclerotic disease (e.g., stroke, **peripheral arterial disease**, ACS) increases risk of ischemic heart disease **(PMHx)**

34(S) Heart-shaped lute with broken string:
A history of atherosclerotic disease (e.g., stroke, peripheral arterial disease, **ACS**) increases risk of ischemic heart disease **(PMHx)**

35(S) Home med bag:
Reviewing the patient's **home medications** can provide clues to comorbidities that increase risk of ischemic heart disease **(Medications)**

36(S) Family of carved wood figures:
a positive **family history of CAD** (i.e., first-degree relative diagnosed prior to age 55 (male) or 65 (female)) is a risk factor for ischemic heart disease **(Family Hx)**

37(S) Tobacco pipe:
Smoking increases risk for ischemic heart disease **(Social Hx)**

39(O) **S**lushi**3** cup:
Listen for extra heart tones (**S3** or S4 gallops) since pre-existing heart disease is a risk factor for angina **(Cardiac Auscultation)**

40(O) **S**tiff **4** chair:
Listen for extra heart tones (S3 or **S4** gallops) since pre-existing heart disease is a risk factor for angina **(Cardiac Auscultation)**

47(P) "Sick Horses Only" sign:
Stress testing is **only indicated in patients with symptoms of angina (2)**

48) Horse running on exercise wheel:
Exercise stress testing uses exercise to increase myocardial oxygen demand

49(P) ".85 x (220 - age)":
Choose exercise stress test if patient is able to reach **85% of maximum predicted HR (220 bpm - age in years) (3)**

50(P) Deconditioned horse:
Pharmacologic stress testing may be used in **patients who cannot exercise** due to limb amputation, arthritis, COPD, deconditioning, etc. **(4)**

51(P) Stimulating spurs on heart:
Pharmacologic stress uses an **inotrope (i.e., dobutamine)** or vasodilator (regadenoson, adenosine, dipyridamole) to stress the heart **(5)**

52(P) Dilated sleeves:
Pharmacologic stress uses an inotrope or **vasodilator (i.e., regadenoson, adenosine, dipyridamole)** to stress the heart **(5)**

53) Fallen black feather on coronary hat:
Vasodilators result in shunting of blood to healthy coronary arteries → **decreased blood flow through diseased, maximally dilated coronary arteries** → ischemia

54) RAD syringe:
Vasodilator stress testing uses **R**egadenoson, **A**denosine, or **D**ipyridamole to increase coronary blood flow

55(P) Heart shield on shirt:
Vasodilators (i.e., regadenoson, adenosine, dipyridamole) can cause **bradycardia** & hypotension **(6)**

56(P) Fainting apprentice:
Vasodilators (i.e., regadenoson, adenosine, dipyridamole) can cause bradycardia & **hypotension (6)**

57(P) Blue face on apprentice:
Avoid vasodilators (i.e., regadenoson, adenosine, dipyridamole) in patients with active COPD or asthma due to risk of **bronchospasm (7)**

58(P) Dropped "X energy" drink:
Patients should **discontinue caffeine, theophylline, & dipyridamole** before undergoing vasodilator stress testing **(8)**

59) Beta-shaped bugle:
Dobutamine (a β-1 agonist) is the inotrope used in pharmacologic stress testing

60) Rapidly shaking heart:
Dobutamine can cause **tachyarrhythmias** and is contraindicated in patients with atrial/ventricular tachyarrhythmias

61) Steam coming out of trampled apprentice's ears:
Dobutamine stress testing is contraindicated in patients with **uncontrolled hypertension (SBP >180 mmHg)**, aortic dissection, or recent ACS (within 1 week)

62) Torn shirt down chest:
Dobutamine stress testing is contraindicated in patients with uncontrolled hypertension (SBP >180 mmHg), **aortic dissection,** or recent ACS (within 1 week)

63) Heart-shaped lute with broken string next to trampled apprentice:
Dobutamine stress testing is contraindicated in patients with uncontrolled hypertension (SBP >180 mmHg), aortic dissection, or **recent ACS** (within 1 week)

64(P) "No outside food or drugs":
Discontinue anti-anginal therapy (e.g., beta blockers, calcium channel blockers, nitrates) prior to diagnostic stress testing **(8)**

65) ECG blanket flying off horse:
Exercise ECG is considered positive if ST depression >1 mm is seen in 1 or more leads

66) Abnormal ECG gate:
Exercise ECG alone cannot be used in patients with baseline ECG abnormalities (e.g., LBBB, ventricular pacing, pre-existing ST depression, digoxin use, or WPW)

67) Echo bullhorn by broken ECG gate:
Exercise stress testing with combined ECG and **echocardiography** can be used in patients with baseline ECG abnormalities

68) ECG gate falling off hinges:
Signs of ischemia on stress echocardiography include **wall-motion abnormalities** not present at rest

69) PET inSPECTor:
Pharmacologic stress **radionuclide myocardial perfusion imaging** is useful if there are baseline wall-motion abnormalities

70) Glowing horseshoe:
Myocardial perfusion imaging uses a **radionuclide tracer** to evaluate blood flow to areas of the heart

71) Purple spot on horseshoe:
Defects that appear on myocardial perfusion imaging after stress but not at rest = areas of myocardial ischemia

77) Short minstrel with short feather cap on NITRO box:
Short-acting nitrates (e.g., nitroglycerin) are first-line treatment for anginal episodes

75) Dilated red shirt sleeves:
Nitrates cause **coronary arterial dilation** AND peripheral VENODILATION → decreases preload and wall stress → decreases myocardial oxygen demand

76) Dilated blue pant legs:
Nitrates cause coronary arterial dilation AND **peripheral VENODILATION** → decreases preload and wall stress → decreases myocardial oxygen demand

78(P) "NITRO **3X**plo**5**ive" box:
Sublingual nitroglycerin can be taken **every 5 minutes for a total of 3 doses (12)**

79(P) "PRO-grade" sign:
Sublingual nitroglycerin may be taken **prophylactically before physical exertion (13)**

81(P) "All are welcome" sash:
Tolerance quickly develops to nitrates → all patients should have a nitrate-free interval every day **(15)**

82(P) "DO NOT overFILL":
Nitrates are **contraindicated in patients taking PDE-5 inhibitors** (e.g., sildenafil, tadalafil) due to risk of hypotension **(16)**

3) Anvil sitting stably on ground:
Chronic **stable angina** = chest discomfort that occurs consistently with a certain level of exertion & is relieved with rest or nitroglycerin

4) Grandfather clock:
Chronic stable angina = chest discomfort that occurs consistently with a certain level of exertion & is relieved with rest or nitroglycerin

13(S) Red wedge on grandfather clock from 2-5:
Each episode of chronic stable angina **usually lasts 2-5 minutes (HPI)**

16(S) Relaxing apprentice:
Ischemic chest pain due to chronic stable angina is typically relieved by **rest** or nitroglycerin **(HPI)**

17(S) Apprentice relaxing on NITRO box:
Ischemic chest pain due to chronic stable angina is typically relieved by rest or **nitroglycerin (HPI)**

41(O) Vitruvian man:
In patients with chronic stable angina, the **physical exam may be normal (Physical Exam)**

42(O) ECG fence with check mark:
Many patients with stable angina have a **normal resting ECG (ECG)**

45) DDx clipboard:
For further workup of Stable Angina, see Chest Pain DDx

46(P) “Stress Test” sign:
In patients with suspected stable angina, **cardiac stress testing** can support the diagnosis and provide risk stratification **(1)**

72(P) “RISK” bag of doctor’s tools:
Stable angina patients with **high-risk features on stress testing** (e.g., myocardial ischemia) should undergo coronary angiography **(9)**

73(P) Minstrel’s beta bugle:
Beta blockers are first line therapy for stable angina **(10)**

74(P) Calci-Yum ice cream drum:
Calcium channel blockers may be used in patients who cannot tolerate beta-blockers or have angina despite beta blocker therapy **(11)**

80(P) Tall minstrel with long feather cap next to NITRO box:
Long-acting nitrates (e.g., isosorbide mononitrate/dinitrate) are second-line therapies for stable angina **(14)**

83(P) “GRANOLA” sacks:
Ranolazine is an option for stable angina patients with refractory symptoms despite optimal therapy **(17)**

84(P) ASA umpire:
Low-dose (81 mg) **aspirin** should be given to patients with stable angina for secondary prevention of cardiovascular disease **(18)**

85(P) Steampunk man:
All patients with stable angina should be on a **statin** for secondary prevention of cardiovascular disease **(19)**

38(S) Cup of cocoa:

Cocaine/amphetamines can cause acute coronary vasospasm **(Social Hx)**

5) Heart-shaped lute with broken string:
Acute coronary syndrome includes unstable angina, NSTEMI, & STEMI

14(S) Red half of wall clock:
Chest pain in **ACS** **usually lasts for 30 minutes** or more **(HPI)**

18(S) Uncomfortable apprentice resting on NITRO box:
Rest and nitroglycerin are less effective at reducing ischemic chest pain due to ACS than with chronic stable angina **(HPI)**

6) Unstable anvil:
Acute coronary syndromes include **unstable angina**, NSTEMI, & STEMI

7) N Stemi ST sign:
Acute coronary syndromes include unstable angina, **NSTEMI**, & STEMI

8) Stemi ST sign:
Acute coronary syndromes include unstable angina, NSTEMI, & **STEMI**

43(O) Q-shaped lasso:
Pathologic Q waves on ECG are an indication of prior ischemia and infarction **(ECG)**

44(O) Cobwebs on ECG fence:
Pathologic Q waves on ECG are an indication of **prior ischemia and infarction** **(ECG)**